Empório da Música

An English-language Python assistant for the fictional musical instrument store in Campo Grande, Brazil.

The support task

Staff repeatedly answer questions about prices, stock, orders and opening hours. Artefact asks for a text agent that reflects the store’s tone, uses the supplied context, chooses the appropriate source and handles questions outside its scope. [Challenge brief, p. 1]

How a customer experiences it

A guest can ask for an acoustic guitar under a budget, select a model and ask for its PIX price in the next turn. The assistant uses short, courteous answers. When asked for cables, it explains the store’s instrument-only range and offers help finding an instrument. [Store manual, p. 2; examples 01, 02 and 05]

How the agent chooses a source

Python recognizes supported phrases and routes prices and stock to catalog tools, order status to an owner-scoped query, and store rules to indexed manual pages. A return question can trigger both an order lookup and a policy search. [Implementation and transcripts below]

Chat router **Tool definitions** **Accessory conversation**

Repository and delivery

The brief requests a public repository with incremental commits, a README explaining setup and technical decisions, and three to five conversations including a non-trivial case. Page 3 of the brief requests the repository URL by email. [Challenge brief, pp. 2–3]

https://github.com/vitorgabriel100/Artefact **Commit history**

02 / DESIGN DECISIONS

How an answer is produced

The brief leaves five technical choices to the candidate. Each decision below describes the implemented behavior and its trade-off. [Challenge brief, pp. 1–2]

Agent approach and policy retrieval

A bounded Python router calls four validated tools. Lexical search expands English terms into Portuguese and returns numbered pages from the eight-page manual. This makes retrieval inspectable, although unfamiliar paraphrases can be missed.

Router **Policy retrieval**

Model, provider and prompt

Optional Ollama generation defaults to llama3.2:3b. A single request includes the customer message, recent exchanges, selected tool evidence and a reference answer. The prompt asks for exact amounts and page references. Generated wording can still introduce errors.

Prompt and HTTP client **Evidence assembly**

Interface and history

The CLI keeps setup small. Within one session, users can refer to a selected product or the last accessible order. Local model mode can use three previous exchanges. Exiting clears this context.

CLI **Session handling**

Data treatment and access

Six CSVs populate in-memory SQLite with relationship checks. Product availability requires active status and positive stock. Money uses integer cents and decimal rounding. Order queries bind both order ID and session customer ID. Reloading each snapshot costs time as data grows.

Catalog and arithmetic **Order access** **Decision record**

What the agent actually said

All five linked transcripts were recorded in offline mode against the supplied data. Each contains the customer messages, full answers and tool arguments. [Challenge brief, p. 2]

01 Find an available guitar under BRL 1,000

The search sets the ceiling to BRL 999.99 and returns five available models. The first is the Tagima Memphis AC-39 at BRL 429.90. [products.csv, categories.csv and promotions.csv]

Read full conversation 1

02 Follow up on the Takamine GD20

Product 95 is active with five units and a table price of BRL 2,199.00. With no active promotion, the 5% PIX discount gives BRL 2,089.05. [products.csv, ID 95; promotions.csv; store manual, pp. 3 and 6]

Read full conversation 2

03 Ask for the address and opening hours

The assistant cites page 2 for Rua 14 de Maio, 3200 and the normal schedule: weekdays 9 AM–6 PM and Saturdays 9 AM–1 PM. Special hours require staff confirmation. [Store manual, p. 2]

Read full conversation 3

04 Decide what can be said about a return

Order 1 is marked delivered. Its estimated delivery date is October 25, 2025. The seven-day online withdrawal period starts at actual receipt. The agent asks for the channel and receipt date before assessing eligibility. [orders.csv, ID 1; store manual, p. 4, §4.1]

Read full conversation 4

05 Resolve “that one” and a request for cables

After selecting the GD20, the customer asks about “that one.” The assistant checks the same model. When the customer asks about cables, it switches to the store’s product-range policy. [products.csv, ID 95; store manual, p. 2]

Read full conversation 5

Run and evaluate the case

From the repository root, use Python 3.11 or later and an activated virtual environment. The offline demonstration runs without a provider key. [README]

```
python -m pip install -e .
python -m unittest discover -s tests -v
python -m emporio.cli chat --mode offline
```

Verification and remaining limits

The final checkpoint installation passed 49 tests on Linux, including CLI replay of all five transcripts. Ollama tests used simulated HTTP responses. Live generation and final Windows validation were pending in the recorded audit. The demo customer ID simulates identity, and stock reflects the supplied snapshot.

Audit **Transcript replay** **Provider tests** **Manual QA**

Development workflow and next steps

OpenAI Codex assisted with code drafts, refactoring, debugging, tests and documentation. Incremental checkpoints expose the decisions for review. Before a production rollout, evaluate paraphrases and generated claims, authenticate customers, connect live inventory and hand unresolved cases to staff.

Setup and AI disclosure **Architecture**

Source guide

Artefact, Desafio Técnico – AI Engineer (supplied brief): p. 1, scenario and required behaviors; p. 2, technical choices, deliverables and examples; p. 3, submission and handling ambiguity. Empório da Música, Manual de Políticas e Procedimentos (supplied manual): p. 2, range, address and hours; pp. 3 and 6, payment and discount rules; p. 4, online returns.

Store manual **Products** **Promotions** **Orders**